

MA378 : Problem Sheet 1

Cian Mac Liatháin

11476078



Q.1 Show that $\sum_{i=0}^n L_i(x) = 1$ for all x .

Interpolating
Polynomial :

$$p(x) = \sum_{i=0}^n f(x_i) L_i(x)$$

for $f(x) = 1$

$$p(x) = \sum_{i=0}^n L_i(x)$$

For all x , $p(x) - f(x) = 1$

$$\therefore \sum_{i=0}^n L_i(x) = 1$$

Q.2 $x_0 = (0, 3)$, $x_1 = (1, 2)$, $x_2 = (2, 4)$

Find Lagrange form of p_2

$$L_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)}$$

$$L_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)}$$

$$L_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)}$$

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x)$$

$$\frac{3(x^2-3x+2)}{2} + \frac{2(x^2-2x)}{-1} + \frac{4(x^2-x)}{2}$$

$$\frac{3x^2}{2} - \frac{9x}{2} + 3 - 2x^2 + 4x + 2x^2 - 2x$$

$$p_2(x) = \frac{3}{2}x^2 - \frac{5}{2}x + 3$$

Q3. $f(x) = \sin\left(\frac{x}{2}\right)$, $x_0 = 0$, $x_1 = \frac{1}{3}$, $x_2 = \frac{2}{3}$, $x_3 = 1$

Cauchy's Theorem: $f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!} \pi_{n+1}(x)$

$$f(x) = \sin\left(\frac{x}{2}\right)$$

$$f'(x) = \frac{1}{2} \cos\left(\frac{x}{2}\right)$$

$$f''(x) = -\frac{1}{4} \sin\left(\frac{x}{2}\right)$$

$$f'''(x) = -\frac{1}{8} \cos\left(\frac{x}{2}\right)$$

$$f^{(4)}(x) = \frac{1}{16} \sin\left(\frac{x}{2}\right)$$

$$\begin{aligned} \pi_4(x) &= (x-0)(x-\frac{1}{3})(x-\frac{2}{3})(x-1) \\ &= x^4 - 2x^3 + \frac{11x^2}{9} - \frac{2x}{9} \end{aligned}$$

$$f(x) - p_3(x) = \frac{f^{(4)}(\tau)}{4!} \pi_4(x)$$

$$f(x) - p_3(x) = \frac{\frac{1}{16} \sin\left(\frac{\tau}{2}\right)}{24} \left(x^4 - 2x^3 + \frac{11x^2}{9} - \frac{2x}{9}\right)$$

Choose $\tau \in [0,1]$ to maximise $\frac{1}{16} \sin\left(\frac{\tau}{2}\right)$
 $\sin(x)$ increases from $0 \rightarrow \frac{1}{2}$

$$\therefore \frac{1}{16} \sin\left(\frac{\tau}{2}\right) \leq \frac{1}{16} \sin\left(\frac{1}{2}\right) \text{ when } \tau \in [0,1]$$

$$|f(x) - p_3(x)| = \frac{\frac{1}{16} \sin\left(\frac{1}{2}\right)}{24} \left|x^4 - 2x^3 + \frac{11x^2}{9} - \frac{2x}{9}\right|$$

$$|f(x) - p_3(x)| = \frac{\sin\left(\frac{1}{2}\right)}{384} \left|\frac{9x^4 - 18x^3 + 11x^2 - 2x}{9}\right|$$

$$\begin{aligned} \left|f\left(\frac{1}{2}\right) - p_3\left(\frac{1}{2}\right)\right| &= \frac{\sin\left(\frac{1}{2}\right)}{384} \left|\frac{9\left(\frac{1}{2}\right)^4 - 18\left(\frac{1}{2}\right)^3 + 11\left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{2}\right)}{9}\right| \\ &= \frac{\sin\left(\frac{1}{2}\right)}{384} \left(\frac{1}{144}\right) \end{aligned}$$

Bound for error of p_3 at $x = \frac{1}{2}$: $0.0000086702 = 8.67 \times 10^{-6}$

Q.4.

$q(x)$ is lagrange polynomial of degree n for $\{(x_i, y_i) : i=0, 1, \dots, n\}$

$r(x)$ is lagrange polynomial of degree n for $\{(x_i, y_i) : i=1, 2, \dots, n+1\}$

$$p(x) = \frac{\overbrace{(x-x_0)r(x)}^{\text{degree } n+1} - \overbrace{(x-x_{n+1})q(x)}^{\text{degree } n+1}}{x_{n+1}-x_0}$$

• $p(x)$ is of degree $n+1$

$$q(x_i) = y_i, \quad \forall i = 0, 1, \dots, n$$

$$r(x_i) = y_i, \quad \forall i = 1, 2, \dots, n+1$$

$$q(x_k) = r(x_k) = y_k \quad \text{for } 1 \leq k \leq n$$

$$\begin{aligned} \bullet p(x_k) &= \frac{(x_k - x_0)r(x_k) - (x_k - x_{n+1})q(x_k)}{x_{n+1} - x_0} \\ &= \frac{(x_k - x_0)y_k - (x_k - x_{n+1})y_k}{x_{n+1} - x_0} = \frac{(x_{n+1} - x_0)y_k}{x_{n+1} - x_0} = y_k \end{aligned}$$

$$\begin{aligned} \bullet p(x_0) &= \frac{(x_0 - x_0)r(x_0) - (x_0 - x_{n+1})q(x_0)}{x_{n+1} - x_0} \\ &= \frac{-(x_0 - x_{n+1})y_0}{x_{n+1} - x_0} \\ &= y_0 \end{aligned}$$

$$\begin{aligned} \bullet p(x_{n+1}) &= \frac{(x_{n+1} - x_0)r(x_{n+1}) - (x_{n+1} - x_{n+1})q(x_{n+1})}{x_{n+1} - x_0} \\ &= y_{n+1} \end{aligned}$$

$p(x)$ is of degree $n+1$ and interpolates the points $\{(x_i, y_i) : i=0, 1, \dots, n+1\}$
 $\therefore$ is the Lagrange polynomial

Q5. Verify that $p_{2n+1}(x) = \sum_{i=0}^n (f(x_i)H_i(x) + f'(x_i)K_i(x))$ solves the HPIP

Conditions to satisfy: degree of $2n+1$, $p_{2n+1} = f(x_i)$, $p'_{2n+1}(x_i) = f'(x_i)$

Recall $L_i(x_k) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$ for $i, k = 0, 1, \dots, n$

$L_i(x)$ is of degree n

$$H_i(x) = \underbrace{[L_i(x)]^2}_{2n} \underbrace{(1 - 2L_i'(x)(x-x_i))}_{+1}$$

• degree of $H_i(x)$ is $2n+1$

$$H_i'(x) = [L_i(x)]^2(-2L_i'(x)) + (1 - 2L_i'(x)(x-x_i)) \cdot 2L_i(x) \cdot L_i'(x)$$

if $i=k$

$$H_i(x_i) = [L_i(x_i)]^2 (1 - 2L_i'(x_i)(x_i - x_i))$$

$$H_i(x_i) = [1]^2 (1 - 0) = 1$$

$$\begin{aligned} H_i'(x_i) &= [L_i(x_i)]^2(-2L_i'(x_i)) + (1 - 2L_i'(x_i)(x_i - x_i)) \cdot 2L_i(x_i) \cdot L_i'(x_i) \\ &= 1^2(-2L_i'(x_i)) + (1 - 2L_i'(x_i)(0)) \cdot 2(1) \cdot L_i'(x_i) \\ &= -2L_i'(x_i) + 2L_i'(x_i) \\ &= 0 \end{aligned}$$

if $i \neq k$

$$\begin{aligned} H_i(x_k) &= [L_i(x_k)]^2 (1 - 2L_i'(x_k)(x_k - x_i)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} H_i'(x_k) &= [L_i(x_k)]^2(-2L_i'(x_k)) + (1 - 2L_i'(x_k)(x_k - x_i)) \cdot 2L_i(x_k) \cdot L_i'(x_k) \\ &= 0 + (1 - 2L_i'(x_k)(x_k - x_i)) \cdot 0 \\ &= 0 \end{aligned}$$

$$H_i(x_k) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$$

$$H_i'(x_k) = 0$$

degree of $H(x)$ and $K(x)$ is $2n+1$

$\therefore \sum_{i=0}^n (f(x_i)H_i(x) + f'(x_i)K_i(x))$ is of degree $2n+1$

also

$$p_{2n+1}(x_i) = \sum_{i=0}^n (f(x_i)H_i(x_i) + f'(x_i)K_i(x_i))$$

$$p_{2n+1}(x_i) = f(x_i)$$

$\therefore$ solves HPIP

$$K_i(x) = \underbrace{[L_i(x)]^2}_{2n} \underbrace{(x-x_i)}_{+1}$$

• degree of $K_i(x)$ is $2n+1$

$$\begin{aligned} K_i'(x) &= [L_i(x)]^2 + x(2L_i(x))L_i'(x) - x_i(2L_i(x))L_i'(x) \\ &= [L_i(x)]^2 + (x-x_i)(2L_i(x)L_i'(x)) \end{aligned}$$

if $i=k$

$$\begin{aligned} K_i(x_i) &= [L_i(x_i)]^2 (x_i - x_i) \\ &= (1)(0) \\ &= 0 \end{aligned}$$

$$\begin{aligned} K_i'(x_i) &= [L_i(x_i)]^2 + (x_i - x_i)(2L_i(x_i)L_i'(x_i)) \\ &= 1^2 + 0 \\ &= 1 \end{aligned}$$

if $i \neq k$

$$\begin{aligned} K_i(x_k) &= [L_i(x_k)]^2 (x_k - x_i) \\ &= 0 \end{aligned}$$

$$\begin{aligned} K_i'(x_k) &= [L_i(x_k)]^2 + (x_k - x_i)(2L_i(x_k)L_i'(x_k)) \\ &= 0 + (x_k - x_i)(0) \\ &= 0 \end{aligned}$$

$$K_i(x_k) = 0$$

$$K_i'(x_k) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$$

Q6. $f(x) = x^5$, $x_0 = 0$ and $x_1 = a$

$$P_3(x) = f(x_0)H_0(x) + f'(x_0)K_0(x) + f(x_1)H_1(x) + f'(x_1)K_1(x)$$

$$f(x_0) = f(0) = 0$$

$$f(x_1) = f(a) = a^5$$

$$f'(x_0) = f'(0) = 0$$

$$f'(x_1) = f'(a) = 5a^4$$

$$\therefore P_3(x) = f(x_1)H_1(x) + f'(x_1)K_1(x)$$

$$L_1(x) = \frac{x-x_0}{x_1-x_0} = \frac{x}{a}$$

$$L_1'(x) = \frac{1}{a}$$

$$H_1(x) = [L_1(x)]^2 (1 - 2L_1'(x)(x-x_1))$$

$$K_1(x) = [L_1(x)]^2 (x-x_1)$$

$$H_1(x) = [L_1(x)]^2 (1 - 2L_1'(x_1)(x-x_1))$$

$$= \left(\frac{x}{a}\right)^2 \left(1 - \left(\frac{2}{a}\right)(x-a)\right)$$

$$= \frac{x^2}{a^2} \left(1 - \frac{2x}{a} + 2\right)$$

$$= \frac{x^2}{a^2} - \frac{2x^3}{a^3} + \frac{2x^2}{a^2}$$

$$= \frac{3x^2}{a^2} - \frac{2x^3}{a^3}$$

$$K_1(x) = [L_1(x)]^2 (x-x_1)$$

$$= \left(\frac{x}{a}\right)^2 (x-a)$$

$$= \frac{x^3}{a^2} (x-a)$$

$$= \frac{x^3}{a^2} - \frac{x^2}{a}$$

$$P_3(x) = f(x_1)H_1(x) + f'(x_1)K_1(x)$$

$$= (a^5) \left(\frac{3x^2}{a^2} - \frac{2x^3}{a^3} \right) + 5a^4 \left(\frac{x^3}{a^2} - \frac{x^2}{a} \right)$$

$$= 3a^3x^2 - 2a^2x^3 + 5a^2x^3 - 5a^3x^2$$

$$= 3a^2x^3 - 2a^3x^2$$